

Developing Mixed Reality Applications with Platform for Situated Intelligence

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ABSTRACT

Both industry and academic interest in mixed reality has skyrocketed in recent years. New headset devices for both virtual and augmented reality are increasingly available and affordable, and new APIs, tools, and frameworks enable developers and researchers to more easily create mixed reality applications. While many tools aim to make it easier to create and interact with content rendered to the headset, these new devices are interesting not just from an output, but also from an input perspective—they contain powerful multimodal sensors that provide unique opportunities to drive forward research on egocentric perception and interaction. In this paper, we introduce Platform for Situated Intelligence—an existing open-source framework—to the mixed reality community. The framework was designed to help developers and researchers create and study real-time, interactive AI systems that process multimodal streaming data. Recent extensions to the framework include new capabilities and components designed specifically to support mixed reality sensory streams and scenarios.

1 INTRODUCTION

Recent years have witnessed an explosion of interest into mixed reality technologies, spanning virtual and augmented reality, across both industry and academia. With potential applications from gaming to education to hybrid work and collaboration, several powerful devices are now on the market for both consumers and researchers, such as Meta's Oculus Quest 2 and Microsoft's HoloLens 2. Several toolkits, libraries, and standardized APIs have been developed to enable the easier creation of new applications and experiences, such as OpenXR [2] (a cross-platform standard providing a unified API for AR and VR), MRTK [1] (a prototyping toolkit integrated with Unity and Unreal), and StereoKit [3] (an easy-to-use open source mixed reality library), just to name a few.

The available tools and frameworks aim to make it easier to create immersive experiences with user interfaces and virtual content rendered to the mixed reality device. Along with rendering, they generally provide basic access to the sensor inputs, e.g., the ability to query the user's head pose or eye gaze. For example, StereoKit makes it easy to create virtual windows, buttons, and sliders, and to access the user's head, hands, and eye poses.

At the same time, given the rich set of multimodal sensors they include, mixed reality devices also present many interesting opportunities for research in egocentric perception and interaction, including complex gesture detection, activity recognition, 3D object tracking, analysis of social group formation and inter-person proxemics, and more. Recent egocentric data collection efforts in the computer vision community reflect an increased interest along these lines [5].

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Tackling such research questions, along with developing mixed reality applications that more deeply leverage the physical context, will require processing and fusing multiple signals, including audio, speech, gaze, IMU, head pose, 3D object positions, etc. Proper synchronization across multimodal data streams is critical for accurate processing and analysis. The various sensor streams produced by mixed reality devices often arrive at different frequencies and latencies. For example, IMU data is generally very high frequency and low latency, whereas more compute-intensive streams such as scene understanding in HoloLens 2 (capturing the presence and locations of floor, wall, and table-top surfaces in the environment) can be produced at lower frequency and with higher latency.

New tools and infrastructures are needed for data fusion, visualization, and annotation, along with an ecosystem of shareable components that can be combined into integrative AI applications. Platform for Situated Intelligence is an existing open-source framework from Microsoft Research that was designed with these goals in mind, targeting the development of interactive multimodal systems that work with streaming data, and for which time and latency are a fundamental concern. Examples include mobile or interactive robots, embodied conversational agents, intelligent meeting rooms, etc.

Although Platform for Situated Intelligence was designed to support a wide array of integrative AI applications, we believe it has great potential for accelerating research and development in mixed reality scenarios, which often use several multimodal sensors on head-mounted devices to enable immersive physically situated experiences. We briefly describe the framework in the next section, and present recently added capabilities and components targeting mixed reality in Section 3.

2 PLATFORM FOR SITUATED INTELLIGENCE

Platform for Situated Intelligence [4] (in short *\psi*) is an open-source¹, cross-platform framework based on .NET for developing real-time, integrative AI systems that work with multimodal streaming data. The framework consists of a *runtime*, a set of *tools* and an open ecosystem of *components*.

The *\psi runtime* provides a model for parallel coordinated computation, where applications are composed of components that communicate with each other via streams of data. Time plays a critical role in the underlying infrastructure: messages flowing through a *\psi* application are timestamped at the source (e.g., sensor) with an *originating time*, and this information is propagated downstream, enabling latency awareness at all points in the application pipeline and supporting a rich set of stream operators for interpolation, synchronization, and data fusion. The availability of these primitives is critical for real-time interactive systems that work with multimodal streaming data, and allows application developers to focus on the task at hand rather than on the lower-level details of coordination, threading, and asynchronous execution. The runtime also provides automatic support for high performance data logging and data replay, simplifying debugging, maintenance, and iterative tuning.

In addition to the streaming infrastructure, the framework also provides *tools* for data visualization, analysis and annotation. Specif-

¹<https://github.com/microsoft/psi>

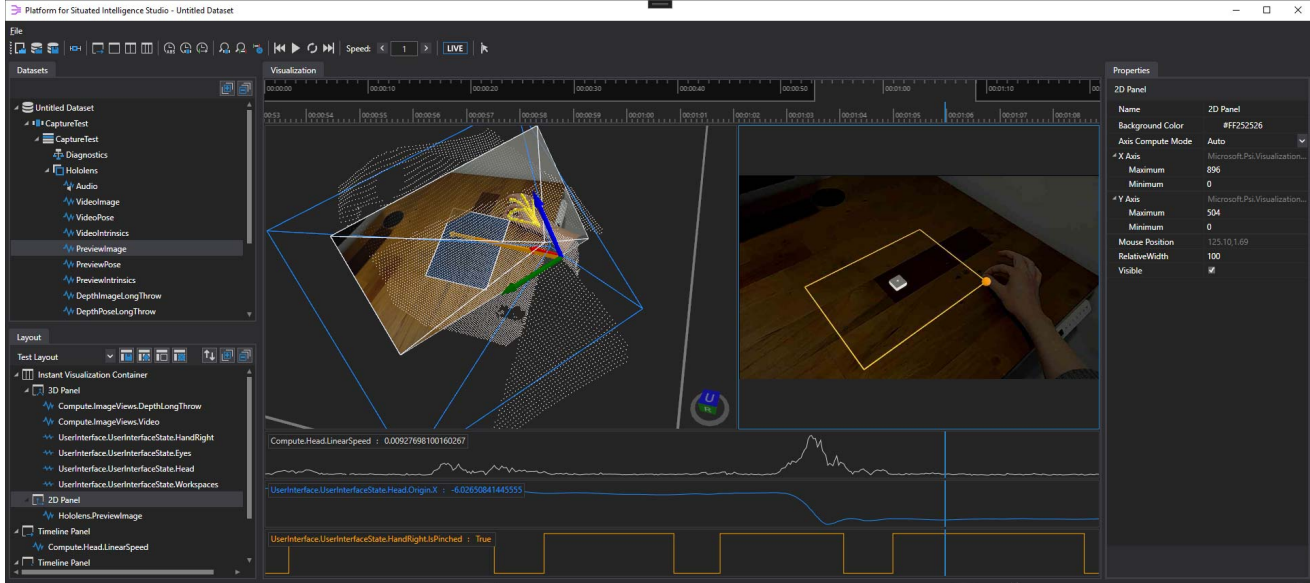


Figure 1: An example layout of custom visualizations for HoloLens 2 data streams in Platform for Situated Intelligence Studio. The top left visualization panel shows an overlay of 3D visualizers for the depth image view (point cloud), projected RGB image view, and tracked hand, head, and gaze, which users can freely pan, rotate, zoom, etc. The top right panel shows a 2D visualizer for the mixed reality preview image, including the RGB camera view along with any holograms that were rendered in the device. The bottom panels show various numerical streams over time.

ically, *Platform for Situated Intelligence Studio* (see Figure 1) is an integrated tool that enables application developers to quickly construct custom visualizations for the data streams persisted by an application. The tool provides an extensive set of visualizers for different data types (e.g., numerical, video, audio, 3D point clouds, etc.) and new visualizers can also be constructed and added. Developers can easily navigate over time and visualize data persisted by an application, either previously collected, or live, as the data is flowing through a running application. The tool also provides diagnostic views of the application pipeline, enables execution of user-defined batch processing tasks, and enables the creation and editing of temporal data annotations.

Finally, *\psi* provides a set of *components* that wrap a variety of technologies, including: sensor components that produce streaming data, such as microphones, cameras, Azure Kinect; multimodal stream processing components for image, audio, speech, etc.; wrappers for executing ONNX deep net models; wrappers for a variety of cloud services; components for audio playback and speech synthesis; and so forth. Components and capabilities also exist for interoperability with other languages and ecosystems, such as Python, JavaScript, and ROS (Robot Operating System) [6]. In addition, developers can build and contribute new components, extending the already existing open ecosystem.

3 MIXED REALITY EXTENSIONS

We have extended Platform for Situated Intelligence with components and visualizers that support the development of mixed reality applications, from capturing and processing the multimodal sensor streams to rendering output on the device. Currently, the extensions primarily target the HoloLens 2 device, but other headsets, such as Oculus Quest, Windows Mixed Reality, Oculus Desktop, SteamVR, etc., may be targeted in the future. The integration between *\psi* and HoloLens 2 is enabled by *StereoKit*—the open-source library mentioned above—and *HoloLens Research Mode* [7]—providing low-level access to key sensors for research applications.

The added components provide easy access to various sensor streams, including depth, RGB and visible light cameras, micro-

phone, head and hand tracking, gaze, IMUs, and scene understanding. Also included are new rendering and spatial sound components. Using *\psi*, developers can easily instantiate and connect these components with other existing components in the ecosystem, or with their own custom components, to create end-to-end applications that leverage the more general stream-processing affordances of the *\psi* framework. The resulting pipelines can be executed directly on the HoloLens 2 device, or may leverage an auxiliary desktop machine for extra processing power and/or disk space for data persistence.

A set of new visualizers enable inspecting and analyzing the data from the various sensors in Platform for Situated Intelligence Studio, as shown in Figure 1. The tool allows developers to scrub through the multimodal data in a synchronized way, with tailored views for 3D (point clouds, gaze direction, hand poses, etc.), 2D (camera images, mixed reality capture, etc.), and timeline data (audio, IMU values, etc.).

4 CONCLUSION

Platform for Situated Intelligence is an open-source framework that was designed to support the development of a wide class of systems characterized by (1) processing of sensory data streams, (2) combining multiple AI technologies, and (3) needing to operate under strict latency constraints (often due to interactivity). Any application that exhibits these characteristics can benefit from the affordances that *\psi* provides, from the infrastructure for representing and reasoning about time, synchronization, and data fusion, to the tools for data analysis, processing, and annotation. Together with the new set of components and visualizers presented in the previous section, we believe *\psi* is well suited for enabling rapid prototyping and accelerating development and research with mixed-reality devices.

To get started, check out our repository of samples², which includes a simple “hello world” app showing the basics of how to use *\psi* with the HoloLens 2. For a deeper dive into the framework’s general architecture, implementation details, and design principles, see the full Platform for Situated Intelligence technical report [4].

²<https://github.com/microsoft/psi-samples>

A number of researchers and developers³ have already been using \psi to build applications in scenarios including multimodal interaction, human-robot interaction, and affective computing, and we hope that with these new extensions and components, the framework can similarly spark the imaginations of developers and researchers in the mixed reality community to create new physically situated applications and experiences.

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³<https://github.com/microsoft/psi/wiki/Who-Is-Using>